

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Summer Examination – 2024**

**Course: B. Tech.      Branch: Mechanical Engineering / Mech. Engg. Sandwich**

**Subject Code & Name: (BTMC503) Theory of Machines- II**

**Semester: V**

**Max Marks: 60**

**Date: 05/07/2024**

**Duration: 3 Hrs.**

**Instructions to the Students:**

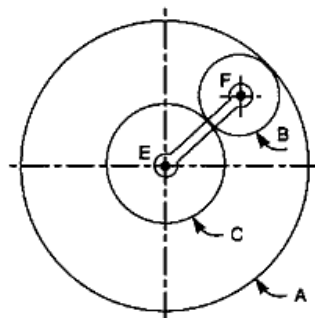
1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (CO) | Marks     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|
| <b>Q.1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |      | <b>12</b> |
| A) Explain with Figure types of Flat belt drives and compare Flat and V-Belt drive.                                                                                                                                                                                                                                                                                                                                                                                  | CO1  | 6         |
| B) Derive an expression for length of Open Belt Drive.                                                                                                                                                                                                                                                                                                                                                                                                               | CO1  | 6         |
| C) Two pulleys, one 450 mm diameter and the other 200 mm diameter are on parallel shafts 1.95 m apart and are connected by a crossed belt. Find the length of the belt required and the angle of contact between the belt and each pulley. How much power can be transmitted by the belt when the larger pulley rotates at 200 rev/min, if the maximum permissible tension in the belt is 1 KN, and the coefficient of friction between the belt and pulley is 0.25? | CO1  | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |      | <b>12</b> |
| A) Define the terms (Any Three) i) Length of Path of contact ii) Length of arc of contact iii) Contact ratio iv) Addendum v) Module                                                                                                                                                                                                                                                                                                                                  | CO2  | 6         |
| B) What is interference in gears? Briefly explain the methods to avoid it.                                                                                                                                                                                                                                                                                                                                                                                           | CO2  | 6         |
| C) A pinion having 17 teeth drives a gear having 49 teeth. The profile of the gears is involute with 20° pressure angle, 6 mm module and Addenda on pinion and gear wheel = 1 module. Find the length of path of contact, arc of contact and the contact ratio.                                                                                                                                                                                                      | CO2  | 6         |
| <b>Q.3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |      | <b>12</b> |
| A) Compare spur gear and helical gear also state their applications.                                                                                                                                                                                                                                                                                                                                                                                                 | CO2  | 6         |
| B) Explain with neat sketch reverted gear train and epicyclic gear train                                                                                                                                                                                                                                                                                                                                                                                             | CO2  | 6         |

- C) An epicyclic gear consists of three gears A, B and C as shown in Fig. The gear A has 72 internal teeth, gear B has 20 external teeth and gear C has 32 external teeth. The gear B meshes with both A and C and is carried on an arm EF which rotates about the center of A at 18 r.p.m in anticlockwise direction. If the gear A is fixed, determine the speed of gear C and its direction of rotation.

CO2

6



**Q.4 Solve Any Two of the following.**

12

- A) Explain in brief,  
 i) Coefficient of fluctuation of energy  
 ii) Coefficient of fluctuation of speed
- B) Explain effect of gyroscopic couple on naval ship during steering.
- C) The turning moment diagram for a multicylinder engine has been drawn to a scale 1 mm = 600 N-m vertically and 1 mm =  $3^\circ$  horizontally. The intercepted areas between the output torque curve and the mean resistance line, taken in order from one end, are as follows: + 52, - 124, + 92, - 140, + 85, - 72 and + 107 mm<sup>2</sup>, when the engine is running at a speed of 600 r. p .m. If the total fluctuation of speed is not to exceed  $\pm 1.5\%$  of the mean, find the necessary mass of the flywheel having radius of gyration of 0.5 m.

CO3

6

CO4

6

CO3

6

**Q. 5 Solve Any Two of the following.**

12

- A) Write short note on longitudinal vibrations, transverse vibrations and torsional vibrations with neat sketch and examples of each.
- B) Derive expression for natural frequency of free torsional vibration of two rotor system
- C) A cantilever shaft 50 mm diameter and 300 mm long has a disc of mass 100 kg at its free end. The Young's modulus for the shaft material is 200 GN/m<sup>2</sup>. Determine the natural frequency of longitudinal and transverse vibrations of the shaft.

CO5

6

CO5

6

CO5

6

\*\*\* End \*\*\*